9 REPRODUCTION IN HUMANS

One of the characteristics of living organisms that makes them different from non-living things is their ability to produce offspring, or **reproduce**. Reproduction is all about an organism passing on its genes. This can be through special sex cells, or gametes. It can also be asexually, without the production of gametes. In this chapter we look at the differences between sexual and asexual reproduction, and study in detail the process of human reproduction.

LEARNING OBJECTIVES

- Understand the differences between sexual and asexual reproduction
- Understand that fertilisation involves the fusion of a male and female gamete to produce a zygote
- Understand that a zygote undergoes cell division and develops into an embryo
- Understand how the structure and function of the human male and female reproductive systems are adapted for their functions
- Describe the role of the placenta in the nutrition of the developing embryo
- Understand how the developing embryo is protected by amniotic fluid
- Understand the roles of oestrogen and testosterone in the development of secondary sexual characteristics
- Understand the roles of oestrogen and progesterone in the menstrual cycle

BIOLOGY ONLY

- Understand the roles of FSH and LH in the menstrual cycle

SEXUAL AND ASEXUAL REPRODUCTION COMPARED

In any method of reproduction, the end result is the production of more organisms of the same species. Humans produce more humans, pea plants produce more pfeea plants and salmonella bacteria produce more salmonella bacteria. However, the way in which they reproduce differs. There are two types of reproduction: **sexual reproduction** and **asexual reproduction**.

In sexual reproduction, specialised sex cells called **gametes** are produced. There are usually two types, a mobile male gamete called a **sperm** and a stationary female gamete called an egg cell or **ovum** (plural **ova**).



▲ Figure 9.1 A sperm fertilising an egg

The sperm must move to the egg and fuse (join) with it. This is called **fertilisation** (Figure 9.1). The single cell formed by fertilisation is called a **zygote**. This cell will divide many times by mitosis to form all the cells of the new animal.

In asexual reproduction, there are no specialised gametes and there is no fertilisation. Instead, cells in one part of the body divide by mitosis to form a structure that breaks away from the parent body and grows into a new organism. Not many animals reproduce in this way. Figure 9.2 shows *Hydra* (a small animal similar to jellyfish) reproducing by budding. Cells in the body wall divide to form a small version of the adult. This eventually breaks off and becomes a free-living *Hydra*. One animal may produce several 'buds' in a short space of time.



▲ Figure 9.2 *Hydra* reproducing asexually by budding

KEY POINT

Individuals produced asexually from the same adult organism are called **clones**.

KEY POINT

A **gene** is a section of DNA that determines a particular characteristic or feature. Genes are found in the nucleus of a cell on the chromosomes (see Chapter 18).

All the offspring produced from *Hydra* buds are genetically identical – they have exactly the same genes. This is because all the cells of the new individual are produced by **mitosis** from just one cell in the body of the adult. When cells divide by mitosis, the new cells that are produced are exact copies of the original cell (see Chapter 17 for a description of mitosis). As a result, all the cells of an organism that are produced asexually have the same genes as the cell that produced them – the original adult cell. So *all* asexually produced offspring from one adult will have the same genes as the cells of the adult. They will *all* be genetic copies of that adult and so will be identical to each other.

Asexual reproduction is useful to a species when the environment in which it lives is relatively stable. If an organism is well adapted to this stable environment, asexual reproduction will produce offspring that are also well adapted. However, if the environment changes significantly, then *all* the individuals will be affected equally by the change. It may be such a dramatic change that none of the individuals are adapted well enough to survive. The species will die out in that area.

SEXUAL REPRODUCTION

There are four stages in any method of sexual reproduction.

- Gametes (sperm and egg cells) are produced.
- The male gamete (sperm) is transferred to the female gamete (egg cell).
- Fertilisation must occur – the sperm fuses with the egg.
- The zygote formed develops into a new individual.

The offspring produced by sexual reproduction show a great deal of genetic variation as a result of both gamete production and fertilisation.

PRODUCTION OF GAMETES

Sperm are produced in the male sex organs – the testes. Eggs are produced in the female sex organs – the ovaries. Both are produced when cells inside these organs divide. These cells do not divide by mitosis but by **meiosis** (see Chapter 17). Meiosis produces cells that are not genetically identical and have only half the number of chromosomes as the original cell.

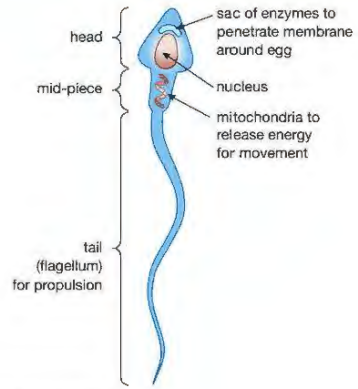
KEY POINT

Cells that have the full number of chromosomes are called **diploid** cells. Cells that only have half the normal number of chromosomes are called **haploid** cells.

TRANSFER OF THE SPERM TO THE EGG

Sperm are specialised for swimming. They have a tail-like **flagellum** that moves them through a fluid. Figure 9.3 shows the structure of a sperm.

Some male animals, such as those of most fish, release their sperm into the water in which they live. The female animals release their eggs into the water and the sperm then swim through the water to fertilise the eggs. This is called *external* fertilisation as it takes place *outside* the body. Before the release takes place, there is usually some mating behaviour to ensure that male and female are in the same place at the same time. This gives the best chance of fertilisation occurring before water currents sweep the sex cells away.



▲ Figure 9.3 The structure of a sperm

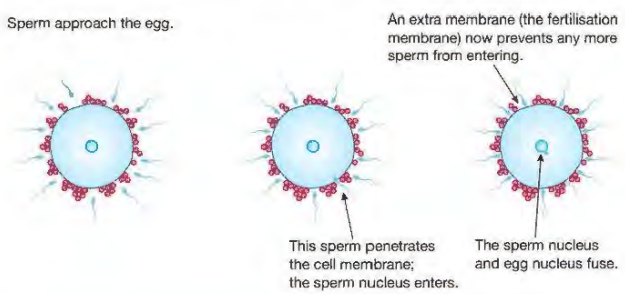
Other male animals, such as those of birds and mammals, **ejaculate** their sperm in a special fluid into the bodies of the females. *Internal* fertilisation then takes place *inside* the female's body. Fertilisation is much more likely as there are no external factors to prevent the sperm from reaching the eggs. Some form of **sexual intercourse** precedes ejaculation.

FERTILISATION

REMINDER

Red blood cells are exceptions. They have no nucleus, so have no chromosomes.

Once the sperm has reached the egg, its nucleus must enter the egg and fuse with the egg nucleus. As each gamete has only half the normal number of chromosomes, the zygote formed by fertilisation will have the full number of chromosomes. In humans, the sperm and egg each have only 23 chromosomes. The zygote has 46 chromosomes, like all other cells in the body. Figure 9.4 shows the main stages in fertilisation.

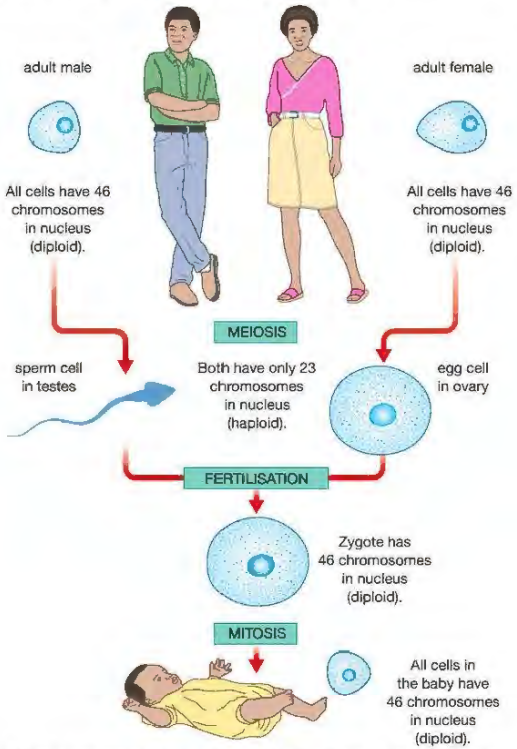


▲ Figure 9.4 The main stages in fertilisation

Fertilisation does more than just restore the diploid chromosome number; it provides an additional source of genetic variation. The sperm and eggs are all genetically different because they are formed by meiosis. Therefore, each time fertilisation takes place, it brings together a different combination of genes.

SUMMARY OF THE HUMAN LIFE CYCLE

Each zygote that is formed must divide to produce all the cells that will make up the adult. All these cells must have the full number of chromosomes, so the zygote divides repeatedly by mitosis. Figure 9.5 shows the importance of meiosis, mitosis and fertilisation in the human life cycle.



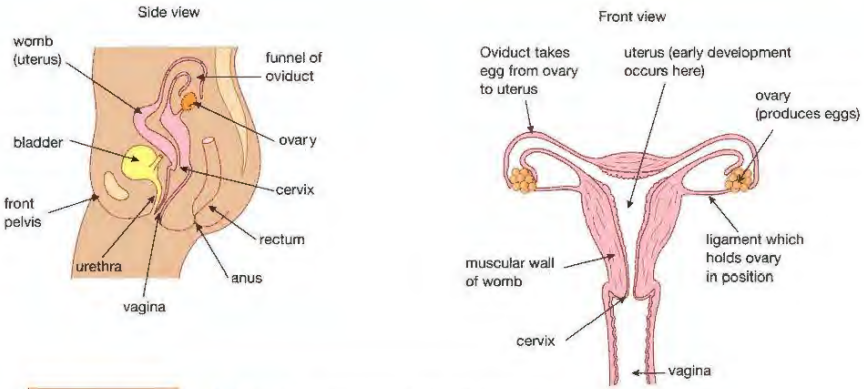
▲ Figure 9.5 The importance of meiosis, mitosis and fertilisation in the human life cycle.

Mitosis is not the only process involved in development, otherwise all that would be produced would be a ball of cells. During the process, cells move around and different shaped structures are formed. Also, different cells specialise to become bone cells, nerve cells, muscle cells, and so on (the process called **differentiation** – see Chapter 1).

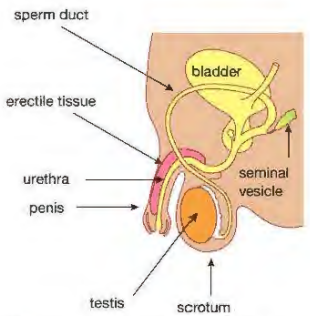
REPRODUCTION IN HUMANS

Humans reproduce sexually and fertilisation is internal. Figures 9.6 and 9.7 show the structure of the human female and male reproductive systems.

The sperm are produced in the testes by meiosis. During sexual intercourse, they pass along the sperm duct and are mixed with a fluid from the seminal vesicles. This mixture, called **semen**, is ejaculated through the urethra into the vagina of the female. The sperm then begin to swim towards the oviducts.

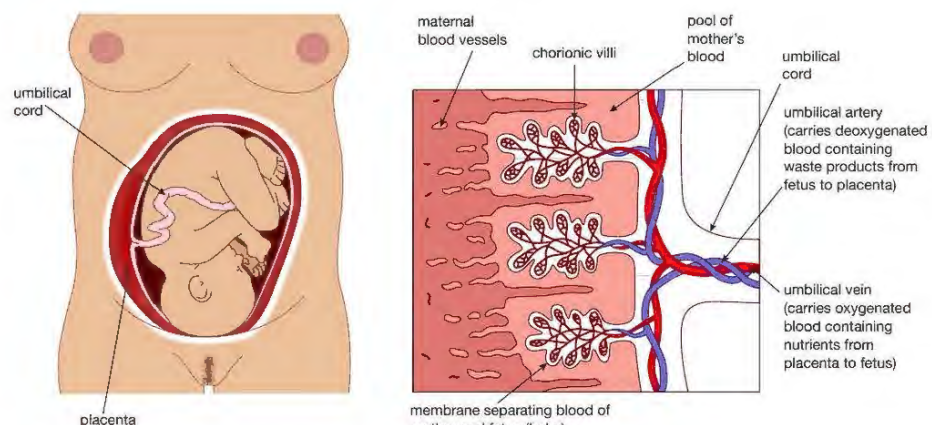


▲ Figure 9.6 The human female reproductive system



▲ Figure 9.7 The human male reproductive system.

Each month, an egg is released into an **oviduct** from one of the ovaries. (The oviduct is also known as the Fallopian tube.) This is called **ovulation**. If an egg is present in the oviduct, then it may be fertilised by sperm introduced during intercourse. The zygote formed will begin to develop into an **embryo**, which will implant in the lining of the uterus. Here, the embryo will develop a **placenta**, which will allow the embryo to obtain materials such as oxygen and nutrients from the mother's blood. It also allows the embryo to get rid of waste products such as urea and carbon dioxide, as well as anchoring the embryo in the uterus. The placenta secretes female hormones, in particular **progesterone**, which maintain the pregnancy and prevent the embryo from aborting (being rejected by the mother's body). Figure 9.8 shows the structure and position of the placenta.



▲ Figure 9.8 The position of the fetus just before birth, and the structure of the placenta.

DID YOU KNOW?

Just before birth, the fetus takes up so much room that many of the mother's organs are moved out of position. The heart is pushed upwards and rotates so that the base points towards the left breast.

During pregnancy, a membrane called the **amnion** encloses the developing embryo. The amnion secretes a fluid called **amniotic fluid**, which protects the developing embryo against sudden movements and bumps. As the embryo develops, it becomes more and more complex. When it becomes recognisably human, we no longer call it an embryo but a **fetus**. At the end of nine months of development, there just isn't any room left for the fetus to grow and it sends a hormonal 'signal' to the mother to begin the birth process. Figure 9.8 also shows the position of a human fetus just before birth.

There are three stages to the birth of a child.

1. **Dilation of the cervix**. The **cervix** is the 'neck' of the uterus. It gets wider to allow the baby to pass through. The muscles of the uterus contract quite strongly and tears the amnion, allowing the amniotic fluid to escape. (In some countries the woman describes this as 'her waters have broken'.)
2. **Delivery of the baby**. Strong contractions of the muscles of the uterus push the baby's head through the cervix and then through the vagina to the outside world.
3. **Delivery of the afterbirth**. After the baby has been born, the uterus continues to contract and pushes the placenta out, together with the membranes that surrounded the baby. These are known as the afterbirth.

Figure 9.9 shows the stages of birth.



1 Baby's head pushes cervix; mucous plug dislodges and waters break.

2 Uterus contracts to push baby out through the vagina.

3 The placenta becomes detached from the wall of the uterus and is expelled through the vagina as the afterbirth.

▲ Figure 9.9 The stages of birth

KEY POINT

Double Award students don't need to study the specifics of FSH or LH. However, you will need to know about other sex hormones, secondary sex characteristics and the menstrual cycle.

DID YOU KNOW?

Sperm production is most efficient at a temperature of about 34 °C, just below the core body temperature (37 °C). This is why the testes are outside the body in the scrotum, where the temperature is a little lower.

HORMONES CONTROLLING REPRODUCTION

Most animals are unable to reproduce when they are young. We say that they are sexually immature. When a baby is born, it is recognisable as a boy or girl by its sex organs.

The presence of male or female sex organs is known as the primary sex characteristics. During their teens, changes happen to boys and girls that lead to sexual maturity. These changes are controlled by hormones, and the time when they happen is called **puberty**. Puberty involves two developments. The first is that the gametes (eggs and sperm) start to mature and be released. The second is that the bodies of both sexes adapt to allow reproduction to take place. These events are started by hormones released by the pituitary gland (see Chapter 7, Table 7.2) called **follicle stimulating hormone** (**FSH**) and **luteinising hormone** (**LH**).

In boys, FSH stimulates sperm production, while LH instructs the testes to secrete the male sex hormone, **testosterone**. Testosterone controls the development of the male **secondary sexual characteristics**. These

include growth of the penis and testes, growth of facial and body hair, muscle development and breaking of the voice (Table 9.1).

In girls, the pituitary hormones control the release of a female sex hormone called **oestrogen**, from the ovaries. Oestrogen produces the female secondary sexual characteristics, such as breast development and the beginning of menstruation ('periods').

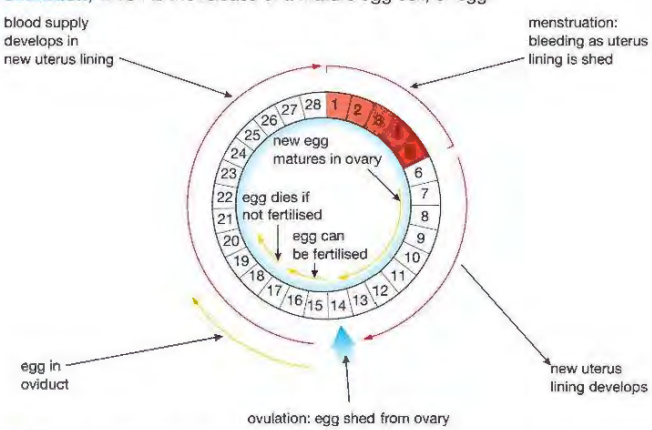
Table 9.1 Changes at puberty.

In boys	In girls
sperm production starts	the menstrual cycle begins, and eggs are released by the ovaries every month
growth and development of male sexual organs	growth and development of female sexual organs
growth of armpit and pubic hair, and chest and facial hair (beard)	growth of armpit and pubic hair
increase in body mass; growth of muscles, e.g. chest	increase in body mass; development of 'rounded' shape to hips
voice breaks	voice deepens without sudden 'breaking'
sexual 'drive' develops	sexual 'drive' develops
	breasts develop

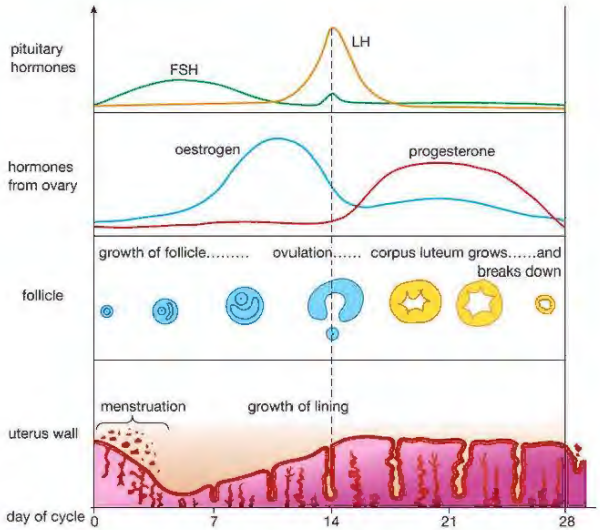
The age when puberty takes place can vary a lot, but it is usually between about 11 and 14 years in girls and 13 and 16 years in boys. It takes several years for puberty to be completed. Some of the most complex changes take place in girls, with the start of menstruation.

HORMONES AND THE MENSTRUAL CYCLE

'Menstrual' means 'monthly', and in most women the **menstrual cycle** takes about a month, although it can vary from as little as two weeks to as long as six weeks (Figures 9.10 and 9.11). In the middle of the cycle is an event called **ovulation**, which is the release of a mature egg cell, or egg.



▲ Figure 9.10 The menstrual cycle



▲ Figure 9.11 Changes taking place during the menstrual cycle

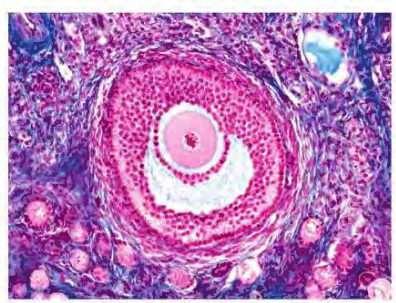
One function of the cycle is to control the development of the lining of the uterus (womb), so that if the egg is fertilised, the lining will be ready to receive the fertilised egg. If the egg is not fertilised, the lining of the uterus is lost from the woman's body as the flow of menstrual blood and cells of the lining, called a **period**.

A cycle is a continuous process, so it doesn't really have a beginning, but the first day of menstruation is usually called day 1.

Inside a woman's ovaries are hundreds of thousands of cells that could develop into mature eggs. Every month, one of these grows inside a ball of cells called a **follicle** (Figure 9.12). This is why the pituitary hormone which switches on the growth of the follicle is called 'follicle stimulating hormone'. At the middle of the cycle (about day 14) the follicle moves towards the edge of the ovary and the egg is released as the follicle bursts open. This is the moment of ovulation.

DID YOU KNOW?

A small percentage of women are able to sense the exact moment that ovulation happens, as the egg bursts out of an ovary.



▲ Figure 9.12 Eggs developing inside the follicles of an ovary. The large follicle contains a fully developed egg ready for ovulation.

While this is going on, the lining of the uterus has been repaired after menstruation, and has thickened. This change is brought about by the hormone oestrogen, which is secreted by the ovaries in response to FSH. Oestrogen also has another job. It slows down production of FSH, while stimulating secretion of LH. It is a peak of LH that causes ovulation.

After the egg has been released, it travels down the oviduct to the uterus. It is here in the oviduct that fertilisation may happen, if sexual intercourse has taken place. What's left of the follicle now forms a structure in the ovary called the **corpus luteum**. The corpus luteum makes another hormone called **progesterone**. Progesterone completes the development of the uterus lining, by thickening and maintaining it, ready for the fertilised egg to sink into it and develop into an embryo. Progesterone also inhibits (prevents) the release of FSH and LH by the pituitary, stopping ovulation.

If the egg is not fertilised, the corpus luteum breaks down and stops making progesterone. The lining of the uterus then passes out through the woman's vagina during menstruation. If, however, the egg is fertilised, the corpus luteum carries on making progesterone, the lining is not shed, and menstruation doesn't happen. The first sign that tells a woman she is pregnant is when her monthly periods stop. Later on in pregnancy, the **placenta** secretes progesterone, taking over the role of the corpus luteum.

EXTENSION WORK

'Corpus luteum' is Latin for 'yellow body'. A corpus luteum appears as a large yellow swelling in an ovary after the egg has been released. The growth of the corpus luteum is under the control of luteinising hormone (LH) from the pituitary.

CHAPTER QUESTIONS

More questions on human reproduction can be found at the end of Unit 2 on page 130.

SKILLS CRITICAL THINKING (7)

1 Which of the following describes sex cells (gametes)?

A diploid cells produced by meiosis

B diploid cells produced by mitosis

C haploid cells produced by meiosis

D haploid cells produced by mitosis

(6)

2 Which of the following is *not* a function of the ovaries?

A the secretion of progesterone

B the production of eggs

C the secretion of oestrogen

D the site of fertilisation

3 Which of the following organs produce(s) the hormone progesterone?

1. pituitary gland
2. ovary
3. uterus
4. placenta

A 1 and 2

B 2 and 3

C 3 and 4

D 2 and 4

SKILLS CRITICAL THINKING (5)

4 A woman's first day of menstruation was on 1st June. Assuming she has a 28-day menstrual cycle, when was she most likely to ovulate?

A 7 June

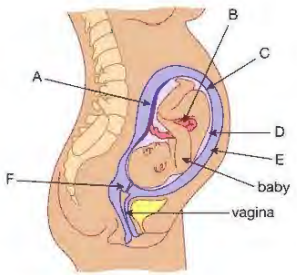
B 10 June

C 14 June

D 21 June

SKILLS ANALYSIS (4)

5 The diagram shows a baby about to be born.



a Name parts A to F on the diagram.

(5)

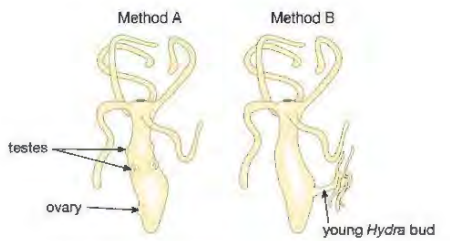
b What is the function of A during pregnancy?

c What must happen to D and E just before birth?

d What must E and F do during birth?

SKILLS CRITICAL THINKING (6)

6 The diagram shows *Hydra* (a small water animal) reproducing in two ways.



a Which of the two methods shows asexual reproduction? Give a reason for your answer.

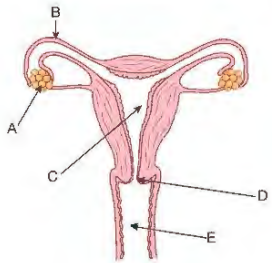
(7)

b Explain why organisms produced asexually are genetically identical to each other and to the organism that produced them.

c When the surroundings do not change for long periods, *Hydra* reproduces mainly asexually. When the conditions change, *Hydra* begins to reproduce sexually. How does this pattern of sexual and asexual reproduction help *Hydra* to survive?

SKILLS ANALYSIS

7 a The diagram shows the female reproductive system.



Which letter represents:

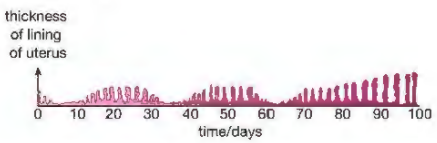
i the site of production of oestrogen and progesterone

ii the structure where fertilisation usually occurs

iii the structure that must dilate when birth commences

iv the structure that releases ova?

b The graph shows the changes in the thickness of the lining of a woman's uterus over 100 days.



SKILLS CRITICAL THINKING

SKILLS INTERPRETATION

i Name the hormone that causes the thickening of the uterine lining.

ii Use the graph to determine the duration of *this* woman's menstrual cycle. Explain how you arrived at your answer.

iii From the graph, deduce the approximate day on which fertilisation leading to pregnancy took place. Explain how you arrived at your answer.

SKILLS CRITICAL THINKING

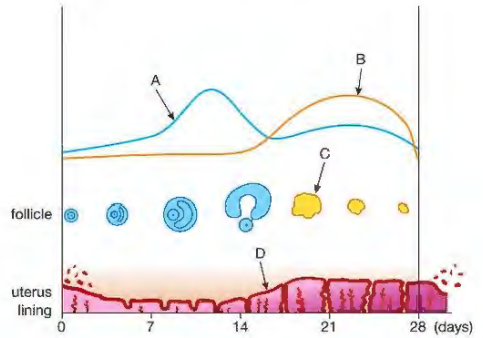
iv Why must the uterus lining remain thickened throughout pregnancy?

SKILLS INTERPERSONAL SKILLS

8 The number of sperm cells per cm^3 of semen (the fluid containing sperm) is called the 'sperm count'. Some scientists believe that over the last 50 years, the sperm counts of adult male humans have decreased. They think that this is caused by a number of factors, including drinking water polluted with oestrogens and other chemicals. Carry out an Internet search to find out the evidence for this. Summarise your findings in about two sides of A4 paper.

SKILLS ANALYSIS

9 The graph shows some of the changes taking place during the menstrual cycle.



a Identify the two hormones produced by the ovary, which are shown by the lines A and B on the graph.

b Name the structure C.

c What is the purpose of the thickening of the uterus lining at D?

d When is sexual intercourse most likely to result in pregnancy: at day 6, 10, 13, 20 or 23?

e Why is it important that the level of progesterone remains high in the blood of a woman during pregnancy? How does her body achieve this:

SKILLS REASONING

i just after she becomes pregnant

ii later on in pregnancy?

BIOLOGY ONLY

SKILLS INTERPRETATION

10 Construct a table of the hormones involved in the menstrual cycle. Use these headings:

Name of hormone	Place where the hormone is made	Function(s) of the hormone

END OF BIOLOGY ONLY